

Conformal Blocks and Crossing Symmetry

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Abstract

Here is a note on a AMAZING topic of 2 dimensional CFT developed in the 80s.

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1 Introduction

1.1 Overview

Since the BPZ paper gives a concrete definition of the conformal blocks, we want to control the behavior of these special functions. A series of papers by Moore and Seiberg in the late 80s and early 90s give an answer. Without concretely calculating the conformal blocks, they show that these function have certain crossing properties, controlled by series of duality matrices. More amazingly, these duality matrices can be explicitly calculated. This gives a explicit way of bootstrapping the 2D CFT structure constants.

Moreover, apart from the field theory application. The crossing properties of conformal blocks forms a beautiful mathematical structure, so called the modular tensor category. This gives us a deep connection between 2D CFT and 3D TQFT. In fact, it can be shown explicitly from the Chern-Simons / WZW correspondence, which is a prototype of holographic duality.

More amazingly, the conformal blocks can also be view as a holomorphic section of a vector bundle over the moduli space of Riemann surfaces. This gives a deep connection between 2D CFT and complex geometry. And in fact this is deeply related to mathematical objects such as modular tensor category and modular functor. This point of view also gives us insights on the quantization of Teichmuller space, which is a key step in understanding AdS3 gravity.

In this note, we will mainly focus on the first two points, which are more directly related to the physics of 2D CFT. (And I don't think I can fully understand the last point, which is more mathematical and abstract.) I'll mainly follow the original papers by Moore and Seiberg.

1.2 Axiomatic CFT

We first review the BPZ axioms of 2D CFT.

1.3 Outline

2 BPZ Conformal Blocks

2.1 Four Point Conformal Blocks

2.2 Torus Conformal Blocks

3 CVO Construction of Conformal Blocks

3.1 Mathematical Reformulation

3.2 Chiral Vertex Operators

The mathematical reformulation is of conceptual importance, yet its not calculational useful.

Bibliography